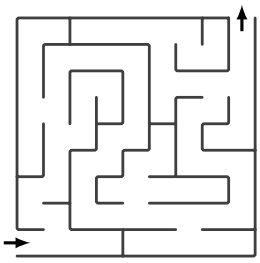
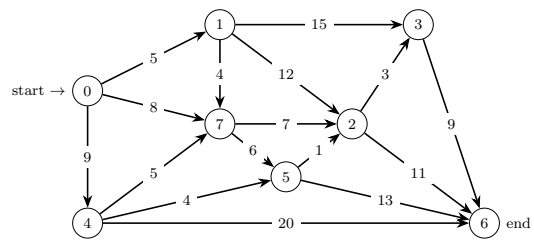
Part IV

Graphs

9 Introduction to Graphs

How do you find a way out of a maze? Or how do you find the shortest path from a starting point to your destination in a one-way road network?





This drawing of a road network is what we call a *graph*. The arrows between various points represent roads. The numbers along the arrows represent the "costs" of traveling along those roads, and the costs can be measurements like distances or times taken to travel along the roads. For example, the cost of reaching point 1 from point 0 is 5.

In toy problems like these, you could trace out a few paths and, perhaps by a bit of trial and error, get to where you need to be. But for larger problems, this wouldn't be practical.

Problems like these – and many others – can be modeled and analyzed with the help of objects known as `Graph`s. Graphs come in a few varieties, each helpful in modeling and solving a particular problem at hand.

9.1 Kinds of Graphs

What's a graph?

A *graph* is a set of points and a collection of lines connecting pairs of points. It is customary to call the points *vertices* or *nodes* and to call the lines *edges*.

More precisely, a graph $G = (V, E)$ consists of a pair of sets:

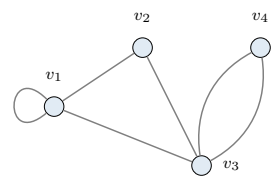
- V: a set of *vertices* (or *nodes*);
- E: a set of *edges*. An edge either connects two vertices with one another or it connects a vertex with itself. An edge connecting a vertex with itself is called a *loop*. The connected vertices are called the *endpoints* of the edge.

Example 41. The following graph $G = (V, E)$ consists of

- the vertex set $V = \{v_1, v_2, v_3, v_4\}$
- the edge set

$$E = \{(v_1, v_1), (v_1, v_2), (v_1, v_3), (v_2, v_3), (v_3, v_4), (v_3, v_4)\}.$$

Graph G with multi-edges and loops
loop: (v_1, v_1); two edges: (v_3, v_4).



pseudographs have loops and multiple edges

This graph is an example of what's called a *pseudograph*. Pseudographs allow both loops and multiple edges between pairs of vertices. A loop is an edge that connects a vertex with itself. In this graph, the edge (v_1, v_1) is a loop, while vertices v_3 and v_4 are connected with two edges.

Many applied problems can be modeled using even simpler graphs than pseudographs. Graphs that do not have "anomalies" like loops or multi edges are called *simple*, and we'll discuss them next.

9.1.1 Simple Graphs

simple graph: no loops, no multiple edges

Simple graphs do not allow loops or multiple edges: you cannot have a vertex connected to itself, and there is at most one edge between any pair of distinct vertices – either the two vertices are connected or they are not. Fig 9.1 shows examples of simple undirected graphs.

9.1.2 Directed vs. Undirected

We distinguish between directed an undirected graphs. When specifying an edge in a directed graph, the order of the vertices matters: the edge (u, v)

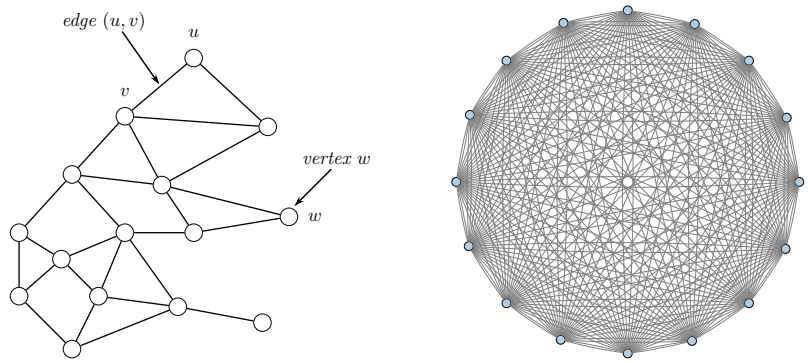


Figure 9.1: Simple undirected graphs can have at most one edge connecting two distinct vertices. The graph on the left has 15 vertices, some pairs of which are not connected. The graph on the right is a simple *complete* graph on 16 vertices, meaning that it has all $\binom{16}{2} = 120$ edges (see also Exercise 9.1).

is different from the edge (v, u). To represent this distinction, we draw an arrow from u to v (instead of drawing a line between u and v) to represent the (u, v) directed edge. For example, Fig 9.2 shows a directed graph that has a (u, v) edge but not a (v, u) edge.

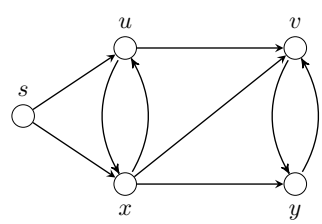
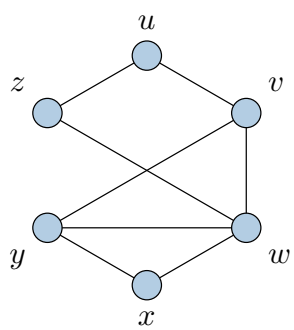


Figure 9.2: A simple directed graph. When specifying an edge in a directed graph, the order of the vertices matters: edge (u, v) is distinct from the edge (v, u). This graph has an (u, v) edge but not a (v, u) edge. It also has both (x, u) and (u, x) edges.

In the rest of this lecture, we are exclusively concerned with undirected simple graphs.

9.2 The Handshake Lemma

The fundamental result in graph theory is known as the *handshake lemma* (which also goes by the name of "the handshake theorem" or "the fundamental theorem of graph theory"). First, we need the notion of the vertex degree.



The *degree* of a vertex is the number of edges incident to it. For example, in this graph, vertex u has degree 2, vertex v has degree 3, and vertex w has degree 4. We write this as

$$\deg(u) = 2, \quad \deg(v) = 3, \quad \deg(w) = 4.$$

The sum of the degrees of all the vertices in any undirected graph is always equal to twice the number of its edges:

$$\sum_{v \in V} \deg(v) = 2\,E. \tag{9.1}$$

For example, the graph shown here has $E = 8$ edges; the sum of its vertex degrees is 16, which is exactly twice the number of edges.

edges = handshakes; $\deg(v)$ = number of handshakes v made; (See Exercise 9.3)

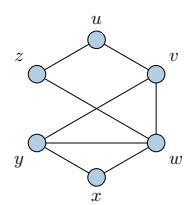
The result in Equation 9.1 is known as the handshake lemma, which also holds for undirected pseudographs, not just simple undirected graphs. We proved this result in class, and Exercise 9.3 should serve as a review of how we proved it.

9.3 Degree Sequences

what is a degree sequence?

A graph's *degree sequence* is the sequence of the degrees of its vertices in nonincreasing order. For example,

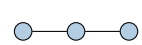
quick example:

the degree sequence of 

is $4, 3, 3, 2, 2, 2$

because one of its vertices, w, has degree 4, two vertices, v and y, each have degree 3, and the other three vertices each have degree 2.

quick example:

Another example: the degree sequence of  is $2, 1, 1$.

a graphic sequence

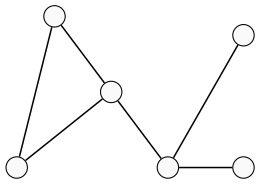
A sequence is called *graphic* if there is a simple graph with that degree

sequence. For example, the sequence $1, 1$ is graphic because it corresponds to the degree sequence of this graph: .

not a graphic sequence

On the other hand, the sequence $2, 1$ is not graphic: no simple graph with two vertices can have a vertex of degree 2.

Example 42. Are the following sequences graphic? (a) $3, 3, 2, 2, 1, 1$; (b) $4, 2, 2, 1, 1, 1$.



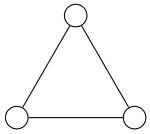
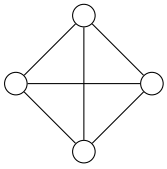
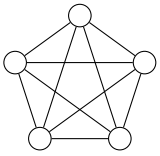
The sequence in (a) is graphic – a graph with that degree sequence is shown on the left.

But the sequence in (b) is not graphic. If such a graph existed, it would have three vertices of an odd degree (of degree 1), which is impossible by the handshake lemma (if you are not sure why, please see Exercise 9.8).

9.4 Some Special Graphs

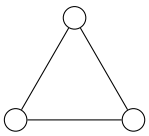
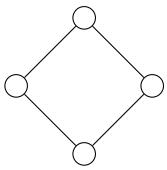
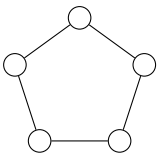
Complete Graphs

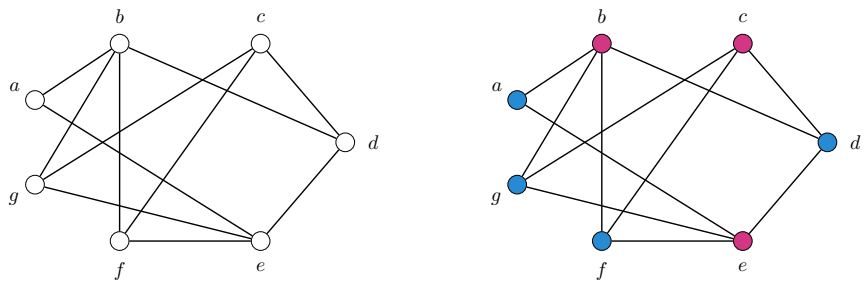
A *complete* graph on n vertices is a simple graph that has an edge between every pair of distinct vertices. It is denoted by K_n. These are the first five complete graphs:

K_1: K_2: K_3:  K_4:  K_5: 

Cycle Graphs

Cycle graphs are defined on $n \geq 3$ vertices. A *cycle* graph on n vertices is denoted by C_n. The number of edges in C_n is also equal to n. We can number the vertices $1, 2, \ldots, n$, so that the n edges are $(1, 2), (2, 3), \ldots (n-1, n), (n, 1)$.

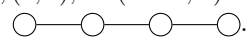
C_3 :  C_4 :  C_5 : 



(a) Graph G. **(b)** G's vertex set is two-colorable.

Figure 9.3: Graph G is bipartite.

Path Graphs

A *path* graph on n vertices has $n-1$ edges. We can identify the vertices as $1, 2, \ldots, n$, so that the $n-1$ edges are $(1,2), (2,3), \ldots (n-1, n)$. Such graphs are denoted by P_n. For example, here's P_4:  .

Bipartite Graphs

The vertices of the simple graph G shown in Figure 9.3a can be split into two disjoint subsets, $V_1 = \{a, d, f, g\}$ and $V_2 = \{b, c, e\}$, so that if (u, v) is an edge in G then $u \in V_1$ and $v \in V_2$.

a bipartite graph Graphs like the one in Figure 9.3 are called *bipartite* or *two-colorable*.

definition A simple $G = (V, E)$ is two-colorable if V can be partitioned into the subsets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 with a vertex in V_2. Thus, if you color the vertices in V_1 with one color and the vertices in V_2 with another, then the endpoints of every edge are of different colors.

Example 43. The cycle graph C_6 is bipartite, as Figure 9.4 shows.

Notice that C_5 is not bipartite (why?). Recall that we proved in class that C_n is bipartite if and only if n is even.

Complete Bipartite Graphs

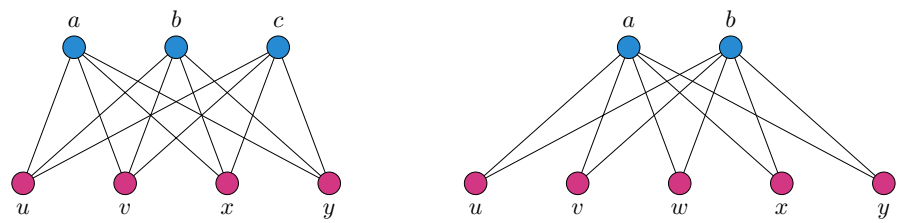
definition: a complete bipartite graph A bipartite graph is called *complete* if there is an edge between every pair of distinct color vertices. The usual notation for a complete bipartite graph on $m+n$ vertices is $K_{m,n}$, where m is the number of vertices colored with one

(a) The cycle graph C_6. **(b)** The vertices of C_6 are two-colorable.

Figure 9.4: C_6 is bipartite.

color and n the number of vertices colored with the other.

Figure 9.5 shows examples of complete bipartite graphs. Notice that the complete bipartite graph $K_{m,n}$ has $m+n$ vertices and $m\,n$ edges (because each of the m vertices is connected to n others and there are no other edges).

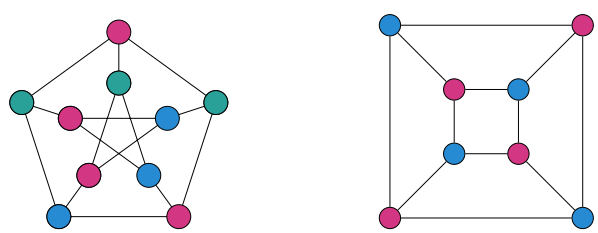


(a) $K_{3,4}$. **(b)** $K_{2,5}$.

Figure 9.5: Examples of complete bipartite graphs.

Regular Graphs

A simple graph is called n-regular if each of its vertices has degree n. For example, both of these graphs are 3-regular.



Note the one on the right is bipartite, while the one on the left isn't. The graph on the left is called Petersen's graph.

9.5 Exercises

9.1. Show that the maximum number of edges a simple undirected graph on n vertices can have is equal to $\frac{n\,(n-1)}{2}$.

9.2. What is the maximum number of edges a simple directed graph on n vertices can have? (Note that loops are still not allowed in simple directed graphs.)

9.3. Prove that in an undirected graph $G = (V, E)$, the sum of all the vertex degrees is equal to twice the number of edges:

$$2\,E = \sum_{v \in V} \deg(v)$$

This result is known as the *Fundamental Theorem of Graph Theory* and goes by the nicknames *"handshake lemma"* and *"handshake theorem."*

Hint: Consider a party where some of the attendees shake hands. At the end of the party, for each attendee, the host records the number of people that the attendee shook hands with. The total tally – the sum of all reported handshakes – is exactly twice the number of handshakes done during the party (because each handshake is reported twice: once by each participant in the handshake). Now model the situation using a graph whose vertices represent the partygoers and edges the handshakes.

9.4. (a) How many edges are there in a graph with 10 vertices each of degree 3? (b) Can the following statement true? Every one of the 17 provinces of Klopstokia borders exactly 3 others.

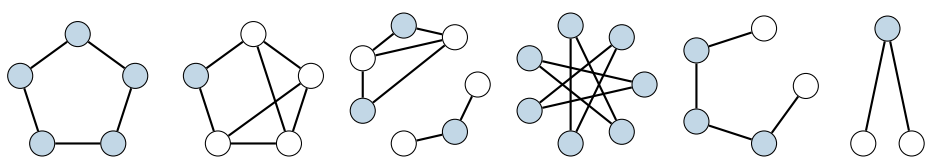
9.5. 127 people play in a singles tennis tournament, prove that at the end of the tournament, the number of people who have played an odd number of games is even.

9.6. Prove that at any party of 31 people, there's always a person who knows an even number of others. (Assume that acquaintance is mutual: if Alice knows Zelda, then Zelda knows Alice.)

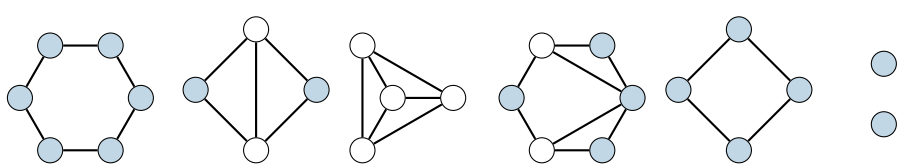
9.7. Prove each of the following claims:

(a) If a graph has an odd number of nodes, then it has a node with even degree.

(b) If a graph has an odd number of nodes, then the number of nodes with even degree is odd.



(c) If a graph has an even number of nodes, then the number of nodes with even degree is even.



9.8. The following claim follows directly from the handshake lemma (Section 9.2, Equation 9.1): "The number of odd-degree vertices is even." Prove this claim.

9.9. In part (b) of Example 42, we gave an example of a non-graphic sequence. Find a few other non-graphic sequences. Challenge yourself and look for non-obvious ones.

9.10. Are the following sequences graphic?

(a) $1, 1, 1, 1, 1$ (b) $3, 3, 3, 3, 2$ (c) $4, 4, 3, 2, 1$ (d) $3, 2, 2, 1, 0$ (e) $5, 4, 3, 2, 1$

9.11. Is the sequence $5, 3, 3, 2, 2, 1$ graphic? If it is, draw the graph; if it isn't, give an argument explaining why.

9.12. Consider this claim: "Every simple graph on $n \geq 2$ vertices contains at least two vertices which have the same degree." Is the claim true? Prove your answer is correct.

Answers to Selected Problems

9.1 A simple undirected graph on n vertices can have at most $\binom{n}{2} = \frac{n\,(n-1)}{2}$ edges. One way to see this is to note that there are $\binom{n}{2}$ pairs of distinct vertices in which the order of the vertices does not matter (that is, the edges (u, v) and (v, u) are indistinguishable). Recall that $\binom{n}{2}$ – pronounced "n-choose-2" – is the standard notation for the number of ways of selecting two distinct elements from a set of n elements. And in a graph, each such selection corresponds to an undirected edge.

If you are not friends with the n-choose-2 notation, no problem. Just think of it this way: Each of the n vertices is connected to each of the other $n-1$ vertices, for a total of $n\,(n-1)$ connections. But this total count double-counts the connections: (u, v) and (v, u) are accounted for as distinct connections in this total count. So if we divide the total count by 2 we get the number of all possible undirected connections. Therefore, the total number of edges in a complete graph on n vertices is $\frac{n\,(n-1)}{2}$.

9.2 Complete graphs – whether directed or undirected – have all possible edges present. So a simple directed graph on n vertices has twice as many edges as a simple undirected graph on n vertices — all $n\,(n-1)$ of them — because each of the n vertices is connected to the other $n-1$ vertices.

9.3 If you think of the partygoers as vertices in a graph and the handshakes as the edges in that graph and complete the argument, the result follows immediately. The degree of a vertex is the number of handshakes the person associated with that vertex reports to the host. The sum of the degrees is the total tally recorded by the host. This number is exactly twice the number of handshakes carried out by the partygoers.

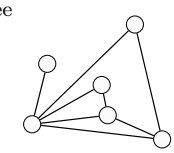
9.7 All of these claims follow directly from the handshake lemma. Make sure you see how. If you do, you should be able to write down your explanation in one or two coherent sentences that will make sense to you when you read what you've written after you've forgotten about it for a few days.

9.8 We proved this in class using the tennis tournament example. The idea was roughly this. First, the even-degree vertices always contribute an even number to the total sum of degrees (no matter how many such vertices are in the graph). Now suppose, by way of contradiction, that the number of odd-degree vertices is odd, making the contribution of the odd-numbered vertices to the total sum of degrees an odd number. But this violates the handshake lemma that says that the total sum of degrees in any graph is always an even number:

an odd number	(odd-deg vertex contribution)
$+$	
an even number	(even-deg vertex contribution)
$\neq$ an even number	(total sum of degrees)

9.9 Many examples are possible. Here are a few impossible degree sequences: (a) $4, 2, 1, 1$ is impossible because such a graph would have 4 vertices with one vertex of degree 4, which is impossible – there are simply not enough vertices for the degree-4 vertex to connect to. (b) Any sequence that has an odd number of odd degrees. For example, $4, 3, 2, 2, 2, 1, 1$ is impossible. (c) Here's another impossible sequence: $6, 6, 6, 4, 4, 4, 1, 1$.

9.11 Yes, this graph has the degree sequence $5, 3, 3, 2, 2, 1$:

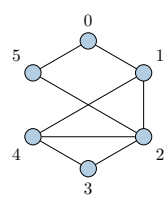


10 Graphs, Continued

10.1 Paths and Cycles in Graphs

If you begin at some vertex and continue moving along the edges of the graph, you are said to be *traversing* the graph. Various traversals go by different names, and those terms can vary depending on which source you're consulting when you read about graphs. But here is the graph traversal vocabulary we'll use:

- A *path* in graph G is a sequence of its vertices connected by edges. The vertices need not be distinct.
- A *simple path* is a path that does not repeat an edge.
- A *circuit* is a path with at least one edge whose first and last vertices are the same.
- A *simple circuit* is a circuit that does not repeat an edge. If all vertices except for the first and last ones along a circuit are distinct, then the circuit is called a *cycle*.

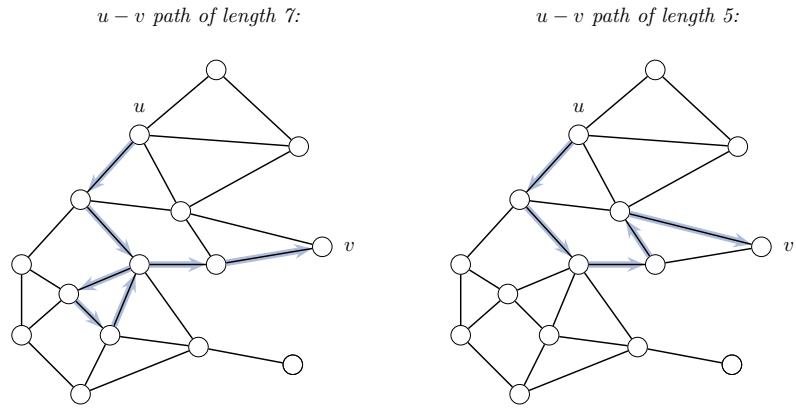


path:	$1-4-2-4-3$	*(repeats edge $2-4$; repeats vertices $2,4$)*
simple path:	$1-2-5-0$	*(no edge is repeated)*
simple path:	$1-2-4-3-2-5-0$	*(repeats vertex 2; no repeated edge)*
circuit:	$1-4-2-3-4-2-1$	*(repeats edge $4-2$)*
simple circuit:	$1-4-2-1$	*(no repeated internal vertex)*

Figure 10.1: Paths and circuits. We use the notation $u-x-y-z-v$ to represent a path from u to v.

Note that if a path does not repeat a vertex, it does not repeat an edge either (but a path not repeating an edge may repeat a vertex).

- The *length* of a path is the number of edges along it:



- A graph is connected if there is a path from any vertex to any other vertex. A graph that is *not connected* partitions into two or more *connected components*.

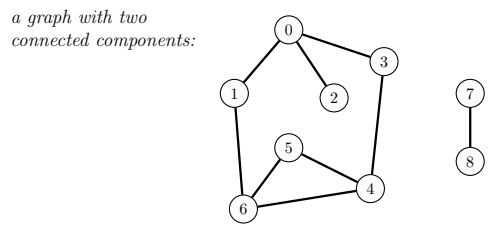


Figure 10.2: A graph with two connected components. Its 9-vertex set $V = \{0, 1, \ldots, 8\}$ is partitioned into two subsets $V_1 = \{0, 1, \ldots, 6\}$ and $V_2 = \{7, 8\}$. You cannot pick two vertices $v_1 \in V_1$ and $v_2 \in V_2$, one from each subset, and find a path between v_1 and v_2.

10.2 Representing Graphs on a Computer

While people may find it convenient to represent graph vertices and edges by drawing dots and lines or arrows between them, computers entertain different preferences when processing graphs.

Of the several ways to store graphs on a computer, we'll find two most applicable to the graph problems we care about: the *adjacency list* and the *adjacency matrix* representations of graphs.

10.2.1 Adjacency List Representation

An array of adjacency lists is a vertex-indexed array of lists; the list stored at index v is a list of v's neighbors, that is, a list of all the vertices connected to v. Fig 10.3 shows the adjacency list representation of an undirected graph G.

Note that each edge appears twice in this representation: If a graph has edge (u, v), then the vertex u appears in v's adjacency list and the vertex v appears in u's list.

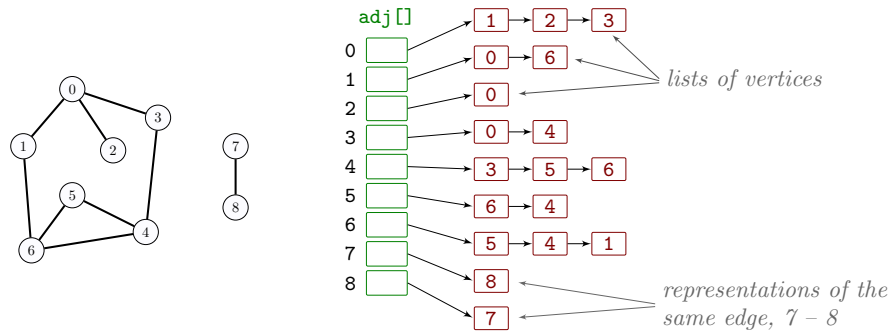


Figure 10.3: The adjacency list array, `adj[]`, stores at index `v` a list of the vertices connected to `v`.

10.2.2 Adjacency Matrix Representation

The adjacency matrix representation keeps a V-by$V$ boolean array shown here as a matrix of 0's and 1's. The entry in row `u` and column `v` is `true` if an only if the vertices `u` and `u` are connected. Fig 10.4 shows the `true` and `false` values as 0's and 1's.

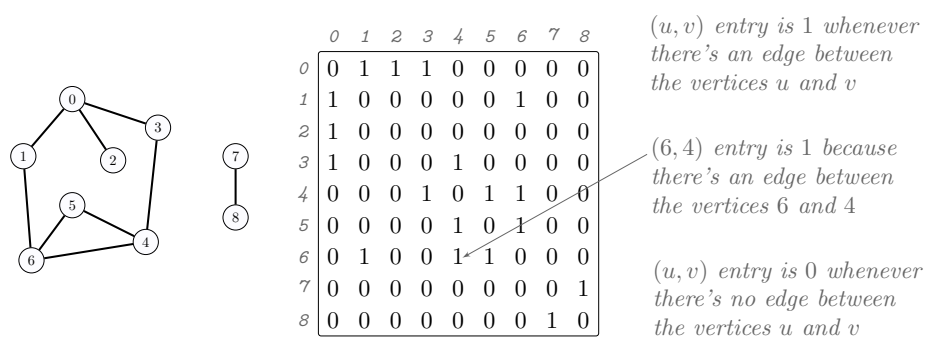
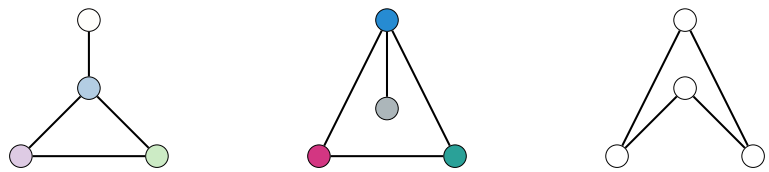


Figure 10.4: An adjacency matrix representation of an undirected graph.

10.3 Graph Isomorphism

When can you say that two graphs are "the same"? You would probably say graph H in Figure 10.5 is different from either F or G, although all three graphs have the same number of vertices and edges.



(a) Graph F. **(b)** Graph G. **(c)** Graph H.

Figure 10.5: Graphs F and G are "the same" while graph H differs from either.

isomorphic = same ("same shape")

So what exactly makes two graphs – like graphs F and G in Figure 10.5 – the same? The idea of "sameness" is captured by the notion of graph *isomorphism*:

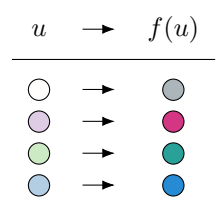
Definition Two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are called isomorphic if there is a one-to-one correspondence f between the vertex sets V_1 and V_2 with the property that

☞ u and v are adjacent in G_1 if and only if $f(u)$ and $f(v)$ are adjacent in G_2.

for all vertices u and v in V_1. The function f is called an *isomorphism* between G_1 and G_2.

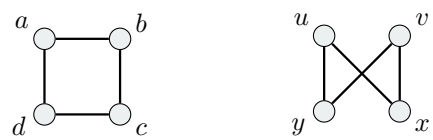
Example 44. Graphs F and G in Figures 10.5a and 10.5b are isomorphic because the following function f maps neighboring vertices in F to neighboring vertices in G and non-neighboring vertices in F to non-neighboring vertices in G:

This f is an isomorphism between F and G of Figures 10.5a and 10.5b:



In class, we did a bunch of examples we took straight from the textbook. As a reminder, they follow below – Examples 45, 46, and 47 are very similar or identical to Examples 9, 10, and 11 on pages 707-708 of your textbook.

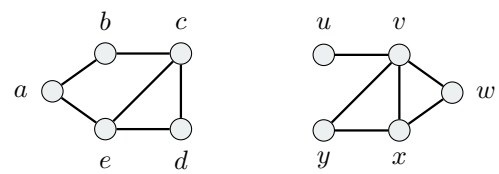
Example 45. Are the following graphs isomorphic?



Answer: Yes, they are. Each graph is the C_4 cycle graph. The following mapping is one possible isomorphism between the two graphs:

$$f(a) = y, \quad f(b) = v, \quad f(c) = x, \quad f(d) = u.$$

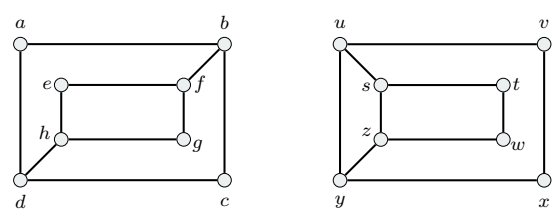
Example 46. Are the following graphs isomorphic?



Answer: No way. One way to see this is to note that the graph on the right has a vertex of degree 1 (vertex u), while the graph on the left doesn't have a vertex of degree 1.

An isomorphism always associates vertices of the same degree with one another. Thus, the *degree sequences* of any two isomorphic graphs *must be identical*. Because the degree sequences of these two graphs differ, the graphs cannot be isomorphic.

Example 47. Are the following graphs isomorphic?



Both graphs have the same degree sequence, $3, 3, 3, 3, 2, 2, 2, 2$. But you can't find an isomorphism between the two.

Answer: One way to see why these graphs are not isomorphic is a follows. Consider vertex a in the graph on the left. Because a's degree is 2, an isomorphism maps a to either v, x, t, or w because those are the only vertices of degree 2 in the graph on the right. But each of v, x, t, or w is adjacent to another vertex of degree two, while the same is not true for vertex a.

10.3.1 Isomorphic Invariants

same number of vertices

Based on the examples discussed so far, you can probably tell that if two graphs are isomorphic then they must share a few properties. For example, isomorphic graphs will clearly have the same number of vertices (because the definition of isomorphism requires a one to one correspondence between the vertex sets of the two graphs).

same number of edges

Similarly, two isomorphic graphs will have the same number of edges (because the definition of isomorphism requires that neighboring vertices map to neighboring vertices and non-neighboring vertices to non-neighboring ones).

Properties shared by isomorphic graphs are known as *isomorphic invariants*.

same degree sequences

Another isomorphic invariant is the degree sequence of a graph. Notice that while isomorphic graphs must have the same degree sequence, the converse isn't true: two graphs can have the same degree sequence and still be non-isomorphic. For example, the two graphs in Example 47 have the same degree sequence but are not isomorphic.

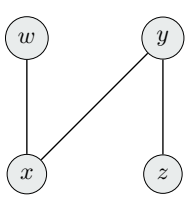
Exercise 10.2 asks you to show that the existence of a simple circuit of length $k \geq 3$ is another graph isomorphism invariant.

Finally, we note that all the conditions for graph isomorphism listed here or mentioned in class are necessary but not sufficient. There are no sufficient conditions for graph isomorphism known today.

10.3.2 Adjacency Matrices and Isomorphisms

one graph, many adjacency matrices

A graph on n vertices can be represented by $n!$ factorial adjacency matrices, each matrix corresponding to the different ordering of the the vertices. For the example, graph G on the left below has $4! = 24$ adjacency matrices, two of which are shown here:



Graph G

$$\begin{array}{c} \\ w \\ x \\ y \\ z \end{array} \begin{array}{c} \begin{array}{cccc} w & x & y & z \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

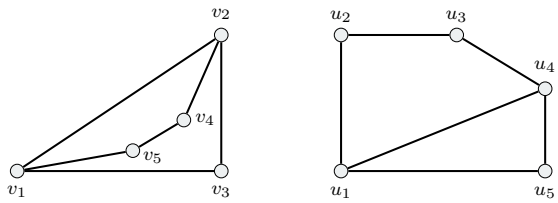
The adjacency matrix for the $w\ x\ y\ z$ vertex order.

$$\begin{array}{c} \\ y \\ w \\ z \\ x \end{array} \begin{array}{c} \begin{array}{cccc} y & w & z & x \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

The adjacency matrix for the $y\ w\ z\ x$ vertex order.

One way to check if graphs G and H are isomorphic is to check if one of G's adjacency matrices is the same as one of H's adjacency matrices.

Example 48. Are the following graphs isomorphic?



Graph G Graph H

Answer: Yes, they are isomorphic. One way to see this is to notice that their adjacency matrices with the following vertex orderings are identical:

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \qquad \begin{array}{c} \\ u_1 \\ u_4 \\ u_5 \\ u_3 \\ u_2 \end{array} \begin{array}{c} \begin{array}{ccccc} u_1 & u_4 & u_5 & u_3 & u_2 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

G's adjacency matrix H's adjacency matrix

Thus, the graphs are isomorphic because they have identical adjacency matrices.

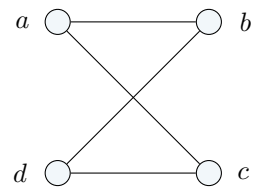
The isomorphic mapping f can be read from the row and column labels of the adjacency matrices. We also show it explicitly in this table on the right.

v	$\longrightarrow$	$u = f(v)$
v_1	$\longrightarrow$	u_1
v_2	$\longrightarrow$	u_4
v_3	$\longrightarrow$	u_5
v_4	$\longrightarrow$	u_3
v_5	$\longrightarrow$	u_2

10.4 Counting Paths

Consider graph G and its adjacency matrix A with respect to the a, b, c, d ordering of its vertices:

$G:$



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

You can use the adjacency matrix to count the number of paths of a given length in G. For example, there are two paths of length 2 from a to d: path $a - b - d$ and path $a - c - d$. Now consider the entry in the a-row and d-column of the square of the adjacency matrix, A^2. That entry is equal to 2.

2 *paths of length* 2 *from* a *to* d.

$$A^2 = \begin{matrix} 2 & 0 & 0 & 2 \\ 0 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{matrix}$$

4 *paths of length* 3 *from* a *to* c.

$$A^3 = \begin{matrix} 0 & 4 & 4 & 0 \\ 4 & 0 & 0 & 4 \\ 4 & 0 & 0 & 4 \\ 0 & 4 & 4 & 0 \end{matrix}$$

8 *paths of length* 4 *from* b *to* b.

$$A^4 = \begin{matrix} 8 & 0 & 0 & 8 \\ 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 \\ 8 & 0 & 0 & 8 \end{matrix}$$

Similarly, the number of paths of length *four* from a to d is the entry in A^4 found in the a's row and d's column. That entry is 8.

This table lists the eight paths of length 4 from a to d.

Paths of length 4 *from* a *to* d*:*

$a - b - a - b - d$
$a - b - a - c - d$
$a - b - d - b - d$
$a - b - d - c - d$
$a - c - a - b - d$
$a - c - a - c - d$
$a - c - d - b - d$
$a - c - d - c - d$

10.4.1 Optional: An Inductive Proof Example

In one of the sections we proved the following theorem using mathematical induction.

Theorem Let G be a graph with adjacency matrix $\mathbf{A}$ with respect to the ordering $v_1, \ldots, v_n$ of vertices (with directed or undirected edges, multiple edges and loops allowed). The number of different paths of length r from v_i to v_j, where $r > 0$ is a positive integer, equals the (i, j)-th entry of $\mathbf{A}^r$.

proof The proof is by mathematical induction on the length of paths, r, between any two vertices. For each $r > 0$, let $P(r)$ be the proposition:

The (i, j)-th entry of $\mathbf{A}^r$ is the number of different paths of length r from v_i to v_j.

Basis Step: (show that $P(1)$ is true) By definition of the adjacency matrix, the number of paths from v_i to v_j of length 1 is the (i, j)-th entry of $\mathbf{A}$. So $P(1)$ is true.

Inductive Step: (show that $P(r)$ implies $P(r + 1)$ for any $r > 0$) For the inductive hypothesis, we assume that that the (i, j)-th entry of $\mathbf{A}^r$ is the number of different paths of length r from v_i to v_j. Because $\mathbf{A}^{r+1} = \mathbf{A}^r\mathbf{A}$, the (i, j)-th entry of $\mathbf{A}^{r+1}$ equals

$$b_{i1}a_{1j} + b_{i2}a_{2j} + \ldots + b_{in}a_{nj}, \qquad \text{(this is the } (i,j)\text{-th entry of } \mathbf{A}^{r+1}\text{)}$$

where b_{ik} is the (i, k)-th entry of $\mathbf{A}^r$. By the inductive hypothesis, b_{ik} is the number of paths of length r from v_i to v_k. A path of length $r + 1$ from v_i to v_j is made up of a path of length r from v_i to some v_k, and an edge from v_k to v_j. The total number of such paths is the product of the number of paths of length r from v_i to v_k (i.e., b_{ik}) and the number of edges from v_k to v_j (i.e, a_{kj}). The sum over all possible intermediate vertices v_k is

$$b_{i1}a_{1j} + b_{i2}a_{2j} + \ldots + b_{in}a_{nj}. \qquad \text{(this is the number of paths of length } r+1\text{)}$$

We have verified that the number of paths of length $r + 1$ from v_i to v_j is the (i, j)-th entry of $\mathbf{A}^{r+1}$, that is, that $P(r)$ implies $P(r + 1)$.

By the principle of mathematical induction, the proposition $P(r)$ is true for all $r > 0$. □

10.5 Summary of Terms

For a simple undirected graph G:

path A *path* in G is a sequence of G's vertices $v_0, v_1, v_2, \ldots, v_n$ such that $e_i = (v_{i-1}, v_i)$ is an edge in G for all $i = 1, 2, \ldots, n$. The vertices need not be distinct.

circuit A *circuit* is a path in G that begins and ends at the same vertex.

simple traversal A path or circuit is *simple* if it does not repeat an edge.

Looking Ahead

In the following lectures, we'll look at special graph traversals. Let's give the definitions now, and we'll talk about them in detail later.

Hamilton traversals A simple path in G that passes through every vertex exactly once is called a *Hamilton path*, and a simple circuit in G that passes through every vertex exactly once is called a *Hamilton circuit*.

Euler traversals An *Euler circuit* in G is a simple circuit containing every edge of G. An *Euler path* in G is a simple path containing every edge of G.

Q What do you mean when you say that two graphs are the same?

A By graphs G and H being "the same" we mean that G and H are isomorphic. The definition of isomorphism is on page 117.

Graphs G and H are isomorphic if it is possible to relabel the vertices of H to produce G. The fact that two graphs are isomorphic does not depend on how the graphs are drawn (if they are drawn at all) or how the vertices of the graphs are labeled.

10.6 Exercises

10.1. The travel cost between two points of a road network is given as an entry in a table shown in Fig 10.6. For example, it costs 4 units to travel between ***b*** and ***c***. If an intersection of a row and a column is empty, then there's no road between the two points. Determine for which one of the tables shown below the following condition is true:

"The smallest cost from ***a*** to ***b*** is at most 6 units."

(The travel cost along any path is the sum of the edge costs along that path.)

(a)

	a	*b*	*c*	*d*	*e*
a				1	
b			4		1
c		4		4	2
d	1		4		
e		1	2		

(b)

	a	*b*	*c*	*d*	*e*
a			3	1	1
b			4		
c	3	4			2
d	1				
e	1		2		

(c)

	a	*b*	*c*	*d*	*e*
a			3	1	
b			4		2
c	3	4			2
d	1				
e		2	2		

(d)

	a	*b*	*c*	*d*	*e*
a			3	1	4
b			4		2
c	3	4			2
d	1				
e	4	2	2		

Figure 10.6: Cost tables for Exercise 10.1. For which one of the tables is it true that the smallest cost from ***a*** to ***b*** is at most 6 units?

10.2. Costs of traversing the edges of a graph on six vertices are given in the matrix of Fig 10.7 (an empty cell indicates the absence of the corresponding edge). What is the least expensive way of getting from a to f?

	a	*b*	*c*	*d*	*e*	*f*
a		2	4			
b	2		1		7	
c	4	1		3	4	
d			3		3	
e		7	4	3		2
f					2	

Figure 10.7: The cost matrix for Exercise 10.2. What's the cheapest way to get from a to f?

10.3. Two independent groups of geology students were asked to chart a network of roads in a remote region.

- Group 1 assigned numbers to the endpoints of the roads, measured the distances (in miles) between the endpoints and summarized the data in the table shown in the left panel of Fig 10.8.

- Group 2 drew the road network on paper and assigned letters $a, b, \ldots, g$ to the points connected by the roads. Their work is shown in the right panel of Fig 10.8.

The problem is, nobody knows how the vertices of Group 2's graph correspond to the vertex numbers used by Group 1. And this is where you are going to help.

Vertex number	1	2	3	4	5	6	7
1				9			7
2				5		11	
3						12	
4	9	5			4	13	15
5				4		10	8
6		11	12	13	10		
7	7			15	8		

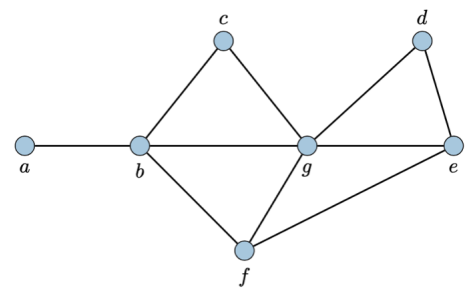


Figure 10.8: The road network for Exercise 10.3. Can we build a correspondence between the numbered vertices shown in the table on the left and the vertices a, b, etc. of the graph on the right?

What is the distance between d and g?

10.4. Two independent groups of geology students were asked to chart a network of roads in a remote region.

- Group 1 assigned numbers to the endpoints of the roads, measured the distances (in miles) between the endpoints and summarized the data in the table shown in the left panel of Fig 10.9.
- Group 2 drew the road network on paper and assigned letters $a, b, \ldots, g$ to the points connected by the roads. Their work is shown in the right panel of Fig 10.9.

The problem is, nobody knows how the vertices of Group 2's graph correspond to the vertex numbers used by Group 1. And this is where you are going to help.

It is known that the length of the shortest path from a to g is at most 15.

What is the distance of the shortest path between e and c?

Vertex number	1	2	3	4	5	6	7
1		11	7	5			12
2	11				13	8	14
3	7			15		10	
4	5		15			9	
5		13				6	
6		8	10	9	6		
7	12	14					

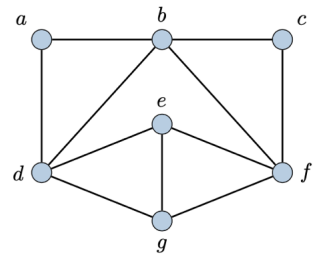


Figure 10.9: The road network for Exercise 10.4. What is the distance of the shortest path between e and c if the length of the shortest path from a to g is at most 15?

10.5. A star in row $1 \leq r \leq 7$ and column $1 \leq c \leq 7$ in the table of Fig 10.10 indicates the presence of a road between r and c. If a star is absent in a cell, then there is no road between the vertices corresponding to that cell.

Determine which numbered vertices in the table in Fig 10.10 can correspond to vertices e and f in the graph.

Vertex number	1	2	3	4	5	6	7
1						*	*
2			*	*		*	
3		*		*			
4		*	*		*		
5				*			*
6	*	*					*
7	*				*	*	

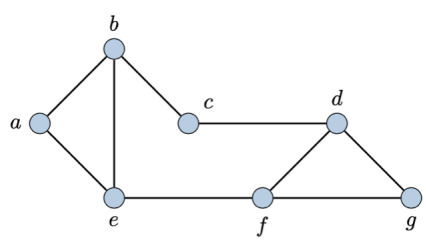
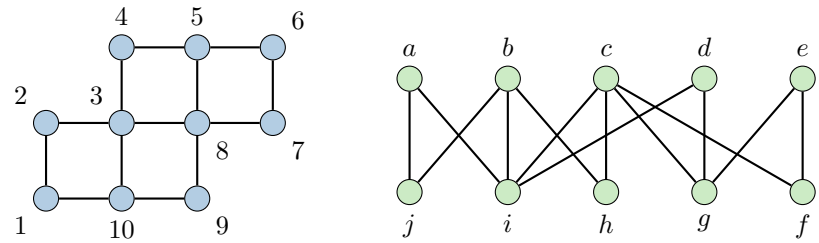


Figure 10.10: The road network for Exercise 10.5. Determine which numbered vertices in the table can correspond to vertices e and f in the graph.

10.6. Why are graphs G and H in Figures 10.5b and 10.5c not isomorphic?

10.7. Are the two graphs in Figure 10.11 isomorphic? If they are, write down the isomorphism. If not, explain why they are not isomorphic.

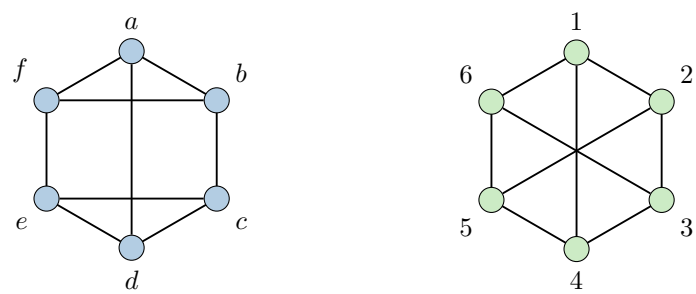


(a) Graph G. **(b)** Graph H.

Figure 10.11: Graphs from Exercise 10.7.

10.8. Show that the existence of a simple circuit of length $k \geq 3$ is an invariant for graph isomorphism.

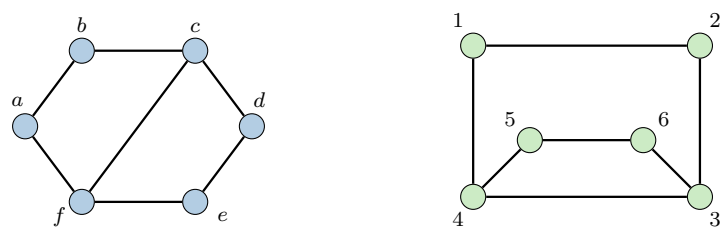
10.9. Are graphs G and H in Figure 10.12 isomorphic? Explain your answer.



(a) Graph G. **(b)** Graph H.

Figure 10.12: Graphs from Exercise 10.9.

10.10. Are the two graphs in Figure 10.13 isomorphic? Explain your answer.



(a) Graph G. **(b)** Graph H.

Figure 10.13: Graphs from Exercise 10.10.

10.11. Is the following *true* or *false*: "If a path in a simple graph does not repeat a vertex, then the path is simple." (The terms path, simple path, and other graph traversal terms are defined on page 114).

10.12. Points a, b, c, d, e, f, and g are connected by a road network.

This table shows the road mileage between connected points. A missing entry in a cell indicates there's no road between the corresponding points.

For example, there is no road between c and b, while the distance between c and d is 2 miles.

What is the shortest distance between a and g?

	a	b	c	d	e	f	g
a		5		12			25
b	5			8			
c				2	4	5	10
d	12	8	2				
e			4				5
f			5				5
g	25		10		5	5	

10.13. Points a, b, c, d, e, f, and z are connected by a one-way road network.

This table shows the road mileage between connected points. A missing entry in a cell indicates there's no road between the corresponding points.

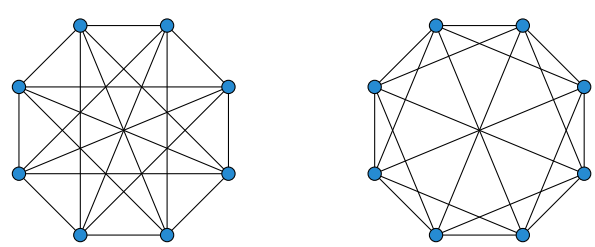
For example, there is a 4 mile long road between a and b, but there's no road between b and a.

How many paths are there from a to z that go through 6 or more points?

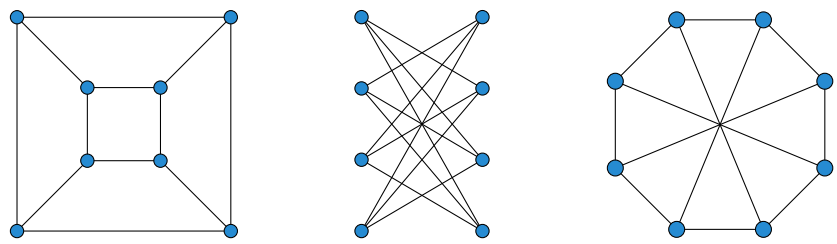
	a	b	c	d	e	f	z
a		7	4				21
b			3				
c				11			13
d					4	12	8
e						4	8
f					5		2
z	66						

Assume that a path never goes through the same point more than once and include a and z in the count of the visited points.

10.14. Are the following two graphs isomorphic?



10.15. Determine which pairs of graphs below are isomorphic.

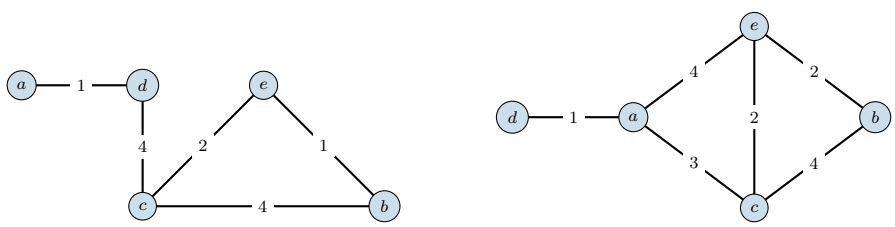


(a) Graph F. **(b)** Graph G. **(c)** Graph H.

Figure 10.14: Which pairs of these graphs, if any, are isomorphic?

Answers to Selected Problems

10.1 The cost matrices given in this problem are small enough, so we can draw these graphs and see which one meets the needed condition. The only table that works is the one shown in $\boxed{\text{(d)}}$.

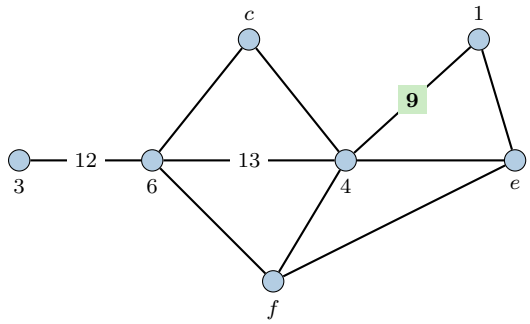


The graph on the left corresponds to the matrix from part(a) and the graph on the right corresponds to the matrix from part (d). In part (a), there's no way to get from a to b for less than 6 units. In part (d) this can be done by going $a \to e \to b$.

10.3 We immediately see that a must correspond to 3 because a is the only vertex of degree 1, and the only matrix row (and column) with a single non-zero entry is the one corresponding to the vertex 3. Then b must correspond to 6 because it's the only vertex connected to a. Similarly, since g has degree 5, it corresponds to 4;

Because c and f are the only other two vertices (other than a and g) reachable from b, we conclude that c and f must correspond to 2 and 5. Thus d can only correspond to 1, which is the only remaining vertex of degree 2.

So we are looking for the distance between the vertices 4 and 1. The table tells us that this distance is $\boxed{9}$.



10.6 Under any isomorphism, a vertex of a given degree must be mapped to a vertex of the same degree. One way to see why G and H are not isomorphic is to notice that graph G has a vertex of degree 1 while graph H doesn't.